

Supplementary File S1 — STROBE / TRIPOD+AI-Style Reporting Checklist

This checklist adapts the **STROBE** statement (observational studies) and the **TRIPOD+AI** checklist (prediction-model reporting) to the cross-dataset machine-learning benchmarking design of this study. Items are mapped to the sections of ``paper_final.md``.

#	Reporting item	Addressed in	Status
1	Study design explicitly described (LODO/LOSO benchmarking)	§1, §3.2	Yes
2	Data sources & eligibility, public repositories named	§3.1, §8.2	Yes
3	Proxy (non-validated) labels explicitly declared & derived	§3.1	Yes
4	Participant-level (not trial-level) partitioning to avoid leakage	§3.2, §4.2	Yes
5	Handling of missing / degenerate data disclosed	§3.5, §4.4	Yes
6	No selective omission of underperforming datasets	§3.5	Yes
7	Predictors / features described (EEG bands, windows)	§2, §3	Yes
8	Model specification (RF, EEGNet, calibration)	§3.3–§3.4	Yes
9	Model performance metrics (Acc, Macro-F1, BAcc, κ) with rationale	§3.2, Table 2	Yes
10	Uncertainty: 20-seed 95% CIs reported	§4.1	Yes
11	Multiple-comparison control : Holm–Bonferroni over 8 targets	§4.1	Yes
12	Significance testing (Wilcoxon signed-rank) described & corrected	§4.1	Yes
13	Model limitations (label collapse, class imbalance) discussed	§4.4–§4.5, §5	Yes
14	Reproducibility: single-script regeneration, fixed seeds	§3.5, §8.1	Yes
15	Code availability (public repo + DOI plan)	§8.1	Yes
16	Data availability (raw accessions + derived intermediates)	§8.2	Yes
17	Ethics / IRB exemption stated	§6	Yes
18	Funding / Competing interests / Author contributions declared	§7 (Author Statements)	Yes

Notes

- STROBE and TRIPOD+AI are reporting frameworks, not study-design mandates;
- All quantitative claims are reproducible from ``run_exp101_reproducible.py``